

Note:
- Instead of using array maybe use classes and attributes
-> should be more performance optimized

--> class: dataset
--> attributes: e.g.
dataset.accelerationX1

- Logging optional depending on uC performance

- All Status changes should be communicated with telemetry

- Instead of Booleans use Status byte and check for status byte

- Some if blocks are secondary and uncritical and are therefore evaluated outside the main branch

Statusbyte structure (should be send to telemetry at the beginning of each loop):

- 1 byte (char)
- States:
 - 0000 0000 (0): Pad Idle
 - 0000 0001 (1): Armed
 - 0000 0010 (2): Liftoff detected
 - 0000 0011 (3): Booster Burnout/Coasting
 - 0000 0100 (4): Apogee detected
 - 0000 0101 (5): Pyro1 signal sent (drogue)
 - 0000 0110 (6): Pyro2 signal sent (drogue)
 - 0000 0111 (7): Pyro3 signal sent (drogue)
 - 0000 1000 (8): Pyro4 signal sent (drogue)
 - 0000 1001 (9): Drogue deployment detected
 - 0000 1010 (10): Pyro1 signal sent (main)
 - 0000 1011 (11): Pyro2 signal sent (main)
 - 0000 1100 (12): Pyro3 signal sent (main)
 - 0000 1101 (13): Pyro4 signal sent (main)
 - 0000 1110 (14): Main deployment detected
 - 0000 1111 (15): Landed
- maybe safe two pyro statuses per parachute (at most two channels per parachute)
 - able to add continuity checks

- IMU:
 - 3 axis acceleration
 - 3 axis angular velocity
 - 2x pressure sensors:
 - barometric pressure values
 - 2x accelerometers:
 - 3 axis acceleration
 - GNSS altitude and velocity (received from sensors)
 - Continuity checks (send to telemetry)
- => Array structure: see size of each datapoint and use predetermined array structure e.g. array[0] to array[3] first IMU integer...

Log sensor data bz sending to sensorics (optional)

acceleration pointing in right direction and cleared of gravity?

0010 (2) [0x2]

Indicating of booster burnout

0011 (3) [0x3]

Apogee detection:

- Change in altitude
 - altitude getting smaller over multiple datasets
- VelZ == 0
- RotationZ > 45°

0100 (4) [0x4]

rotation threshold up to debate
-> indicated rocket is pitching which (ideally) only happens at apogee

Two of the following:

- 0101 (5) [0x5]
- 0110 (6) [0x6]
- 0111 (7) [0x7]
- 1000 (8) [0x8]

Send down via telemetry

Altitude < Main deploy setpoint?

Two of the following:

- 1010 (10) [0xA]
- 1011 (11) [0xB]
- 1100 (12) [0xC]
- 1101 (13) [0xD]

Send down via telemetry

Done by reading photoresistors/distance sensor
-> distance threshold for the latter

1110 (14) [0xE]

Received CMD to send status?

Send Status to telemetry

No more changes to altitude & z-acceleration?

1111 (15) [0xF]

Send down via telemetry

